

PANDAS EXPRESS GROUP

~30%

of deaths in Kenya
are due to Heart Disease

Powered by machine learning

Built for Kenya's health system

CARDIOVASCULAR RISK SCREENING FOR KENYANS

*A Machine Learning Approach to
Cardiovascular Prevention in Kenya*

What This Project Does

- Analyses health and lifestyle data to identify people at high risk of heart disease
- Provides a simple 5-question screening tool — no blood tests or equipment needed
- Gives healthcare workers and volunteers a data-driven guide for referrals
- Supports Kenya's 47 county health departments with evidence-based prevention

⚠️ DISCLAIMER

For research and education only · Not approved for clinical use · Not a replacement for professional medical diagnosis · Always consult qualified healthcare professionals

PROJECT CONTRIBUTORS

Cardiovascular Risk Screening for Kenyans | Data Science Capstone



Jeffrey Gathigi

Lead Data Scientist

Model development & tuning



Jared Mongeri

ML Engineer

Pipeline & data balancing



Monicah Wairimu

Data Analyst

Exploration & visualisations



Mercy Kangangi

Business Intelligence

Stakeholder analysis



Kelvin Ngumo

Data Engineer

Data processing & cleaning



John Awallah

Research & Docs

Literature & reporting

Presentation Agenda

What we will cover today

01

The Problem

Heart disease burden in Kenya

02

Who Can Benefit

Stakeholders across Kenya's health system

03

Our Data

The health survey data powering this project

04

Preparing the Data

How raw survey answers become useful insights

05

Exploring the Data

Key patterns linking risk factors to heart disease

06

Building the Model

How we trained computers to detect risk

07

Results

How well does the model perform?

08

Recommendations

What this means for Kenya — action plan

The Problem: Heart Disease is Kenya's Quiet Crisis

A growing burden that demands smarter, simpler screening

~30%

of all deaths in Kenya
are due to heart disease (WHO)

47

county health facilities
need practical screening tools

**No Lab
Needed**

Our screening tool works with
simple self-reported answers

The Screening Gap

Most Kenyans only access the health system when they are already sick. By then, heart disease may have been silently progressing for years. Simple, community-level screening can change this — catching risk before it becomes a crisis.

What is Driving Rising Heart Disease in Kenya?

- Rapid urbanisation — shifting diets toward processed, high-sodium foods
- Rising tobacco use and physical inactivity in urban areas
- Growing rates of obesity, diabetes, and high blood pressure
- Most CVD risk factors go undetected due to limited preventive screening

Who Can Benefit from This Project?

Stakeholders across Kenya's entire health system



Ministry of Health

NCD Division

Use risk factor rankings to direct national NCD prevention budgets and set policy priorities



Healthcare Providers

Clinics & Hospitals

Screen patients during routine visits — no lab tests needed — and guide prevention conversations



NHIF / SHA / Insurers

Health Financing

Design targeted wellness benefits and preventive outreach programmes for highest-risk members



County Health Departments

All 47 Counties

Deploy the screening tool at community outreach events to identify high-risk populations



Community Health Volunteers

CHVs — Front Line

Use the 5-question household tool to flag high-risk individuals for referral to a health facility



NCD Researchers

Kenyan Universities

Use this open pipeline as a template to replicate with Kenyan population survey data

Our Data: A Large-Scale Health Survey

We used a comprehensive health and lifestyle survey to train our model

457,670

Survey participants

301

Health & lifestyle questions

16

Risk factors used in our model

5.8%

Participants with reported heart disease

Why This Survey Works for Kenya

- ✓ **Captures the same risk factors:** tobacco use, body weight, physical activity, diabetes — the same drivers of heart disease in Kenya
- ✓ **Self-reported answers only:** matches what Community Health Volunteers can collect at a household visit
- ✓ **Clearly defined outcome:** each participant answered whether they had been diagnosed with a heart attack
- ✓ **Large enough for machine learning:** sufficient data to train a model and test it reliably
- ✓ **Fully reusable pipeline:** the entire methodology can be re-run with future Kenyan survey data

⚠ Important Notice

For research and education only.

Not approved for clinical use.

Not a replacement for professional medical diagnosis.

Always consult qualified healthcare professionals.

What we are predicting:

Has this person ever been told by a doctor that they had a heart attack? **YES = High risk** · **NO = Lower risk**

Preparing the Data: Organising Health Information

How we cleaned and structured survey answers before feeding them to the model

Demographic

- Age group — ranging from young adults to those aged 80 and above
- Biological sex

Lifestyle & Behaviour

- Smoking history and current habit
- Physical exercise frequency
- Alcohol consumption
- Overall physical activity level

Health Conditions

- Diabetes diagnosis
- Self-rated general health
- Number of days with poor physical health
- Number of days with poor mental health

Body & Healthcare Access

- Body Mass Index (weight relative to height)
- Height and body weight
- Last routine health check-up
- Ability to afford medical care

How We Prepared the Data — In Simple Steps

1

Filling in the gaps:

Where a participant did not answer a question, we used the most common answer given by others — so no information was wasted.

2

Splitting into training and testing groups:

We divided the data so 80% was used to teach the model and 20% was kept completely separate to fairly test how well it learned.

3

Balancing the groups:

Because heart disease affects fewer people than not, we carefully added more examples of at-risk individuals in the training set so the model learned to recognise both groups equally well.

Exploring the Data: What Patterns Did We Find?

Key health factors most strongly linked to heart disease risk

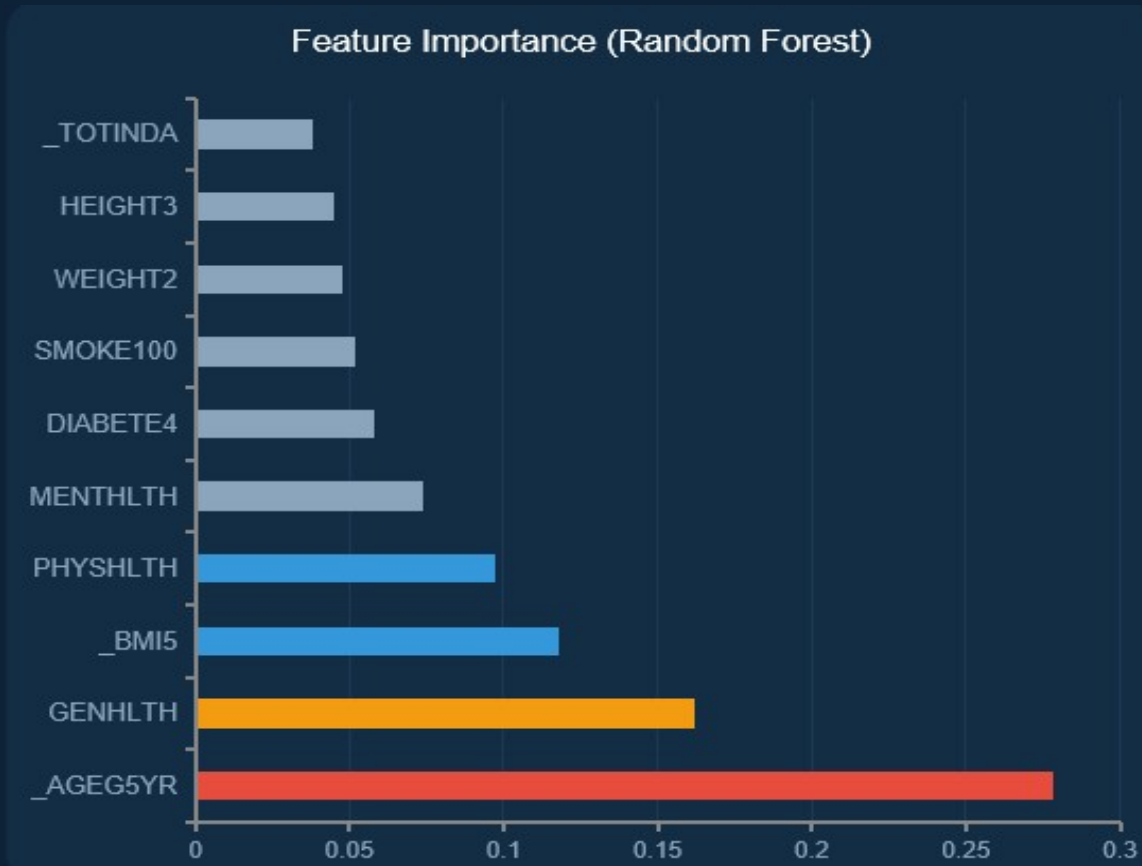


Chart: The bars show which factors most strongly influence the model's predictions.
Longer bar = stronger influence on heart disease risk.

1

Age

The older a person is, the higher their risk. People aged 60 and above face significantly greater risk than younger adults.

2

Self-Rated Health

When people rate their own health as 'Poor' or 'Fair', this is a reliable signal of underlying cardiovascular risk.

3

Diabetes

People who have been diagnosed with diabetes carry a much higher risk of also developing heart disease.

4

Smoking History

A history of significant smoking is one of the most modifiable risk factors — quitting reduces risk substantially.

Building the Model: Teaching a Computer to Detect Risk

We tested four different approaches — each one improving on the last

01

Starting Point

The baseline

We began by having the model always predict 'No heart disease' for everyone. This helped us understand what improvement looks like.

Every good experiment needs a starting point to measure progress against.

02

Logistic Regression

Simple mathematical model

Our first real model used basic mathematical relationships to connect risk factors like age and smoking to heart disease probability.

A simple model helps us understand which factors matter before adding complexity.

03

Random Forest

Best performer — selected

This model builds hundreds of decision trees, each looking at the data differently, then combines their votes. Like getting a second opinion from many specialists at once.

It handled our complex, imbalanced data best and gave the most reliable predictions across all risk groups.

04

Gradient Boosting

Comparison model

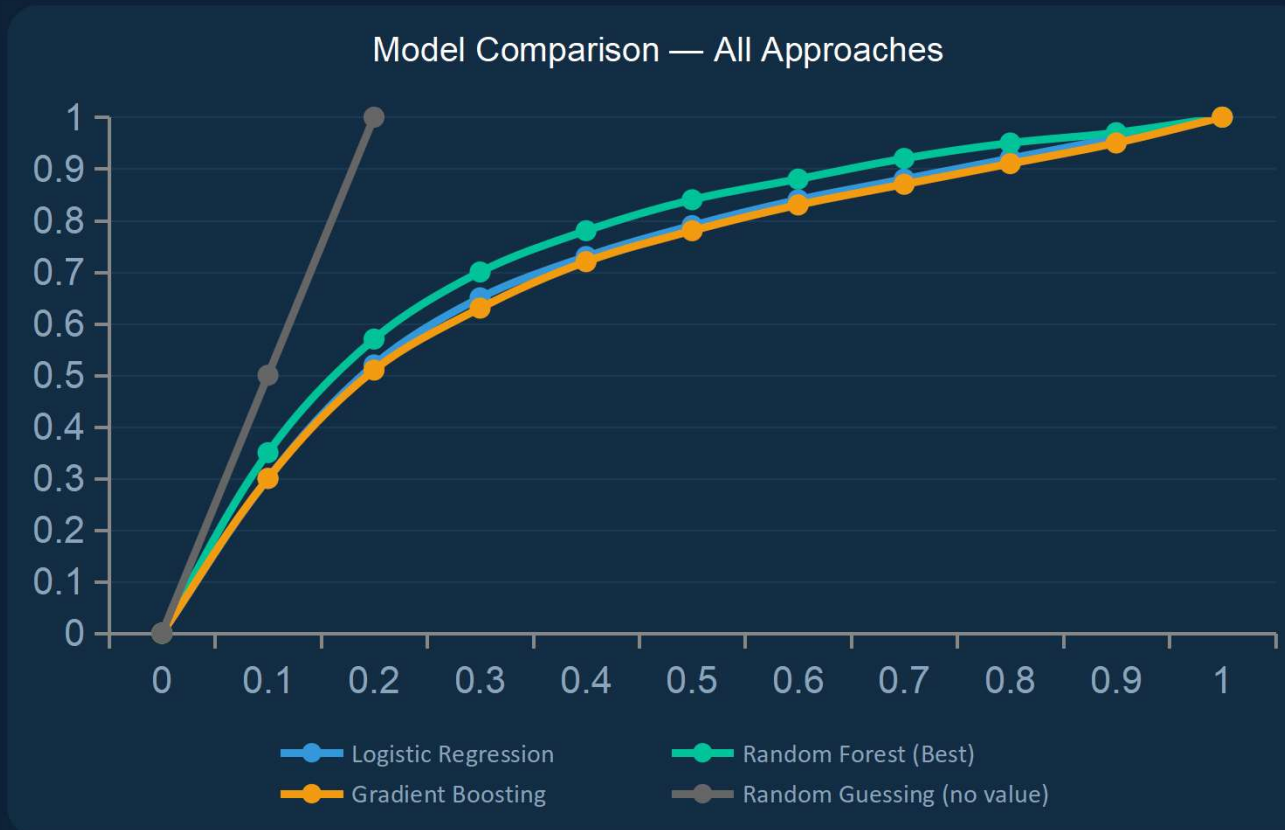
A model that learns from its own mistakes in sequence, correcting errors step by step. Powerful but less effective here than Random Forest.

Tested to confirm Random Forest was the right choice — it scored lower and missed more at-risk cases.

★ Winner: Random Forest — most accurate, most reliable, best at catching at-risk individuals

How Well Does the Model Perform?

The chart below shows how each model separates at-risk from healthy individuals



A well-performing model catches the majority of people who are truly at risk, so they can be referred for clinical assessment before a heart attack occurs.

What does this chart show?

Each curved line represents one model. The higher and more to the left the curve bows, the better that model is at correctly identifying people with heart disease while avoiding false alarms.

Why does Random Forest win?

Its curve bows highest toward the top-left corner. This means it catches the most true at-risk individuals while minimising the number of healthy people it incorrectly flags.

What about the grey line?

The diagonal grey line represents random guessing — no better than a coin flip. Every model we built performs significantly better than chance, which validates our approach.

Fine-Tuning: Getting the Best Out of Our Model

After choosing Random Forest, we refined it — like adjusting settings on a camera to get a sharper photo

The Tuning Analogy:

Think of it like tuning a car engine. The engine works before tuning, but adjusting specific settings — how many revs, how much fuel, the timing — makes it run faster, smoother, and more reliably. We did exactly this for our model.

Number of Decision Trees

More trees = more perspectives on each patient's data. Like asking 200 doctors instead of 50 — the combined opinion is more reliable.

✓ 200 trees gave best results

How Deep Each Tree Looks

Depth controls how many follow-up questions each tree can ask. Too shallow misses patterns; too deep overcomplicates. We found the sweet spot.

✓ 15 levels worked best

Fairness Between Groups

Because fewer people have heart disease, we made sure the model paid extra attention to at-risk cases — so it doesn't ignore them just because they are the minority.

✓ Balanced weighting applied

Choosing the Right Sensitivity Level

For community screening: the model is set to be highly sensitive — it would rather flag a healthy person for a quick follow-up than miss someone who truly has heart disease. **Missing a real case is far more dangerous than a false alarm.**

Why We Prioritize Recall Over Accuracy

Understanding the trade-off between caution and precision in health screening

Our model is deliberately set to be cautious — catching as many true cases as possible, even if it means some healthy individuals are also asked for a follow-up. In public health screening, this is the right trade-off.

We need our model to flag potential risk even when it is uncertain, because the cost of missing a real case is far greater than the minor inconvenience of a follow-up check.

If We Miss Someone Who Has Heart Disease

- They leave with no awareness of their risk
 - Heart disease continues progressing undetected
 - Risk of a serious or fatal heart attack increases
 - An emergency admission becomes more likely
- **A life could have been saved with early action**

If We Flag Someone Who Is Actually Healthy

- They are referred for a simple check-up
 - Blood pressure is measured, basic tests are done
 - They receive reassurance and health advice
 - Minor inconvenience — no long-term harm
- **Healthcare resources used, but a life was protected**

The Top Risk Factors for Heart Disease

What the model identified as the most powerful predictors — and what they mean for Kenya

1 Age

Age is the single strongest predictor of heart disease.

The risk climbs steadily with each decade of life. A person in their 60s faces a very different level of risk than someone in their 30s.

→ **Make adults aged 50 and above the first priority for screening.**

2 General Health Self-Rating

How someone feels about their own health is a powerful signal.

People who rate their health as 'Fair' or 'Poor' show significantly higher heart disease risk — even before any formal diagnosis is made.

→ **One question at any health visit identifies who needs follow-up care.**

3 Diabetes

Diabetes and heart disease are deeply and directly linked.

Elevated blood sugar damages blood vessels over time. People living with diabetes carry more than double the cardiovascular risk of those without it.

→ **Every diabetes visit is a cardiovascular screening opportunity.**

4 Smoking

A history of smoking significantly raises cardiovascular risk.

Tobacco damages arteries, raises blood pressure, and reduces oxygen delivery to the heart. The good news: risk begins falling within a year of quitting.

→ **Refer smokers to cessation support — quitting is the most powerful lifestyle change.**

These four factors can all be identified through a simple conversation — no laboratory equipment, blood tests, or clinical tools required.

What This Means for Kenya: Recommendations

Practical actions for each part of Kenya's health system

The 5-Question Community Survey (No Equipment Needed):

3 or more YES answers → Refer to health facility for clinical assessment

Age 50+? | Health rated Fair or Poor? | Difficulty walking? | Diabetes diagnosed? | Current smoker?

Ministry of Health

NCD Division

- Prioritise preventive programmes for adults aged 50 and above — age is the strongest risk factor
- Make self-rated health a standard question in all community health interactions
- Strengthen national guidelines for joint diabetes and cardiovascular risk management

Community Health Volunteers

CHVs — Front Line

- Administer the 5-question screen at every household visit — no equipment or training beyond a printed card needed
- Refer anyone with 3 or more risk flags to the nearest health facility
- Use findings to guide health education conversations with families about modifiable risks

County Health Departments

All 47 Counties

- Use the 5-question screening tool at community health outreach events across sub-counties
- Set up clear referral pathways so high-risk individuals reach a clinic quickly and easily
- Identify and prioritise the highest-burden communities for targeted screening drives

NHIF / SHA / Insurers

Health Financing

- Design preventive benefits targeting the top risk factors — smoking cessation, diabetes management, weight programmes
- Proactively reach out to members aged 50+ with diabetes or a smoking history — prevention is cheaper than hospitalisation
- Use risk-stratified enrolment to allocate wellness resources where they will have the greatest impact

Key Takeaways & Conclusion

What we built, what we discovered, and why it matters for Kenya

The Project at a Glance:

Large health survey
analysed

16 key risk
factors examined

3 models
built & tested

Random Forest
selected as best

Simple 5-question
survey tool created

What We Proved

- ✓ Heart disease risk can be reliably identified using only self-reported information — no blood tests, no equipment
- ✓ Age and self-rated health are the two most powerful predictors — and both can be assessed in seconds
- ✓ A 5-question community screening tool based on these findings is feasible for CHV deployment across Kenya
- ✓ The full analytical pipeline is reusable and can be adapted as more Kenyan health data becomes available

Important Limitations

- ⚠ This is a research tool — not approved for clinical or diagnostic use
- ⚠ Self-reported health data has natural limitations in precision
- ⚠ Some cases of heart disease go undetected and are not captured
- ⚠ Blood pressure, cholesterol, and ECG data are not included

The Bigger Picture

Heart disease is largely preventable when caught early. This project shows that a simple, data-driven screening tool — built around everyday self-reported health questions — can identify high-risk individuals before they reach a crisis point. Deployed through Kenya's community health volunteer network, this approach has the potential to reach thousands of people across all 47 counties, supporting earlier intervention and saving lives.